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UNIVERSITI
TEKNOLOGI
MARA

**FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES
UiTM SHAH ALAM, SELANGOR**

**BACHELOR OF INFORMATION SYSTEM (HONS.)
INFORMATION SYSTEM ENGINEERING**

**CSC584 ENTERPRISE PROGRAMMING
GROUP PROJECT**

TITLE: Uni-Run: Marathon & Virtual Run Coordinator

PREPARED BY:

GROUP	STUDENT ID	NAME
CDCS2664B	2024206986	NIK AMIRUL HAQIMI BIN NIK ZAHIRUDDIN
CDCS2664B	2024242906	MUHAMMAD ALIFF AIMAN BIN ROHAIZAD
CDCS2664B	2024433848	AMIR HARITH BIN MUHD SHAHRUNNIZAM
CDCS2664B	2024405276	MUHAMMAD ALIF HAKIMI BIN MOHD FAUZI
CDCS2664B	2025422298	AMMAR MUKHRIZ BIN MOHAMMAD SAAD

PREPARED FOR:

SIR MUHAMAD RIDHWAN BIN MOHAMAD RAZALI

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INTRODUCTION

The more the campus marathons and virtual running events are gaining popularity, the more need there is for an effective and central event management system. Manual registration and participant management, event scheduling, and results compilation can lead to delays, data inaccuracies, and administrative burden. However, as more people join events, traditional methods start to fall short in supporting efficient coordination and a seamless user experience. But as events get larger, the traditional techniques get less effective in coordinating events and providing a good user experience.

To solve these problems, this project presents Uni-Run: Marathon & Virtual Run Coordinator, a web-based application for an easier running event to manage, both on campus and virtually. The system offers a centralized platform for participants to register for events, check their registration status, and see their assigned bib numbers and finish times. Meanwhile, administrators can oversee event information, approve participant sign-ups, allocate bib numbers, monitor race outcomes and track event performance overall with an interactive dashboard.

The system is implemented with Model-View-Controller (MVC) architecture using Java Enterprise Web Technologies, in which the Model is implemented by using JavaBeans, the View is implemented by using JSP, and the Controller is implemented by using Servlets. This design is known for its security in session management for user authentication and system access, as well as making code more organized, maintainable, and scalable.

PROBLEM STATEMENT

1. Inefficiencies in Manual Event Coordination and Tracking

Current methods of organizing campus marathons and virtual runs often rely on fragmented, manual processes or generic spreadsheets. This leads to administrative bottlenecks, such as delays in verifying registrations, errors in assigning official bib numbers, and difficulties in manually tracking and updating runners' final finish times. Without a centralized system, managing data across both physical and virtual event formats becomes highly error-prone and labor-intensive.

2. Lack of Centralized and Real-Time Event Analytics

Event coordinators face significant challenges in gathering and analyzing participant metrics effectively. There is a lack of automated tools to calculate community-wide achievements, such as total cumulative distance covered, or to visualize real-time demographic breakdowns. This absence of a dynamic data visualization dashboard hinders organizers from gaining actionable insights to improve future event engagement and marketing strategies.

3. Security Vulnerabilities and Poor Session Management in Registration Platforms

Many entry-level web portals for community events lack robust architectural design, making them vulnerable to security risks during participant data entry. Without a secure, well-structured framework and strict session management, the processes for participant login, registration, and logout remain exposed to session hijacking or data mishandling. This compromises user privacy and undermines trust in the platform's reliability.

OBJECTIVE

The primary purpose of this project is to develop the Uni-Run system to streamline the coordination of campus and virtual running events. Specifically, the project aims to achieve the following objectives:

1. To implement a secure, MVC-based web application architecture:

By utilizing Java Web Technology, this objective involves structuring the system into distinct Model (JavaBean), View (JSP), and Controller (Servlet) components. Additionally, it includes establishing robust session management to ensure secure participant registration, login, and logout processes.

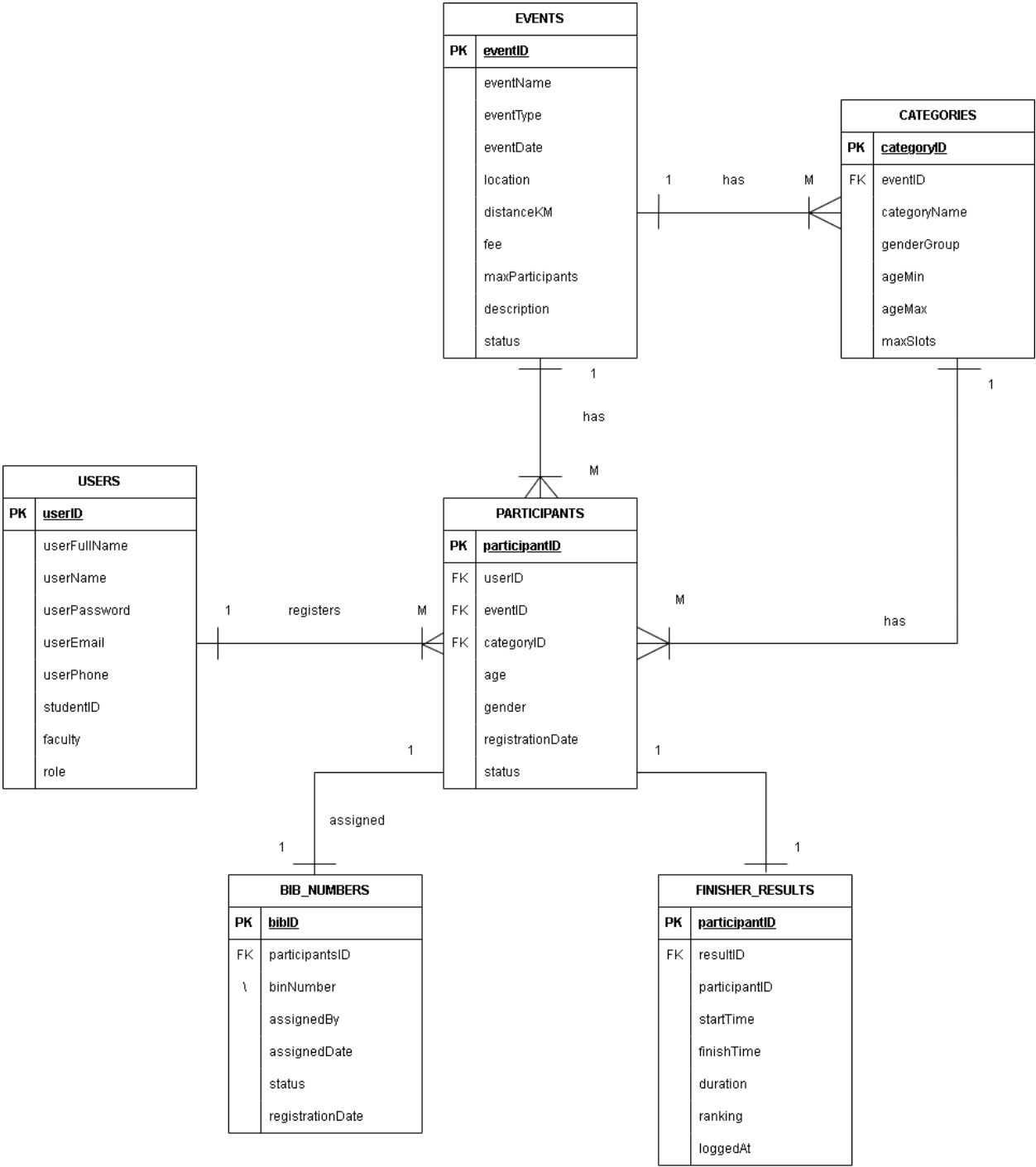
2. To streamline event registration and data management processes:

This objective focuses on developing a comprehensive CRUD module that allows participants to easily sign up for upcoming campus and virtual runs. It also empowers system administrators with the necessary tools to update event statuses, assign official bib numbers, and accurately record runners' final finish times.

3. To design an interactive dashboard for visualizing event metrics:

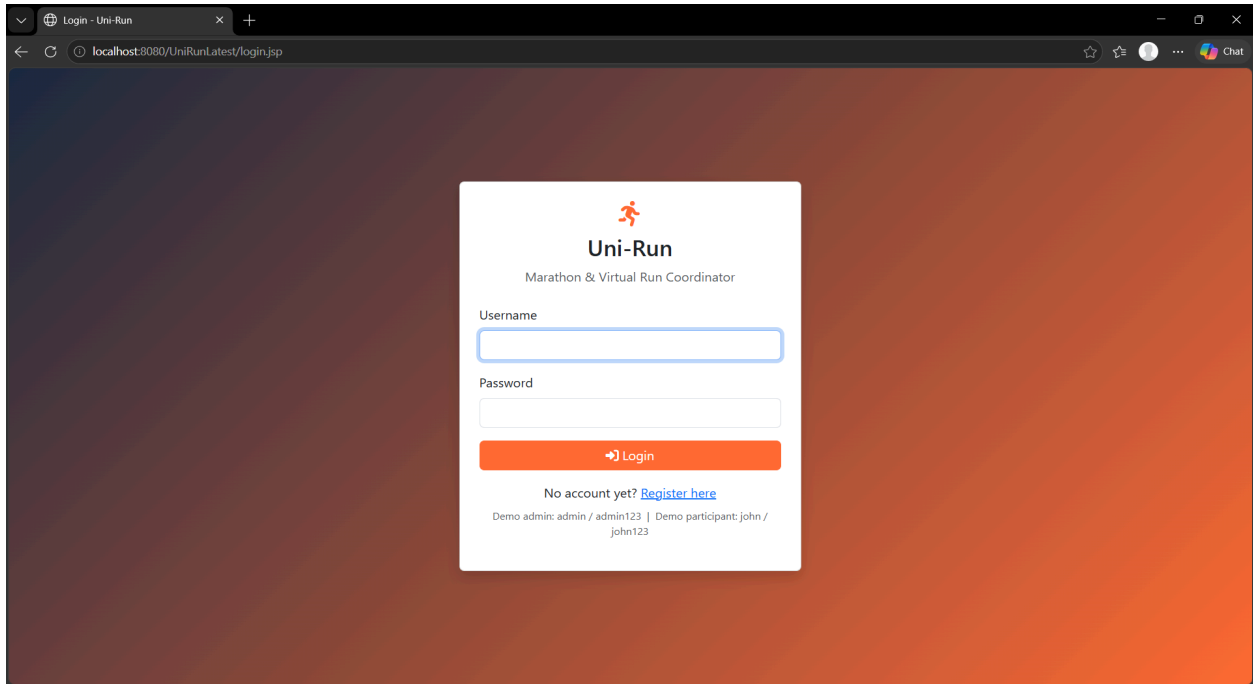
This involves creating a dynamic user interface that provides a clear, real-time overview of key statistics and participant data. The dashboard will specifically calculate and display the total cumulative distance covered by the community, alongside detailed demographic breakdowns of participants categorized by age and gender.

ERD



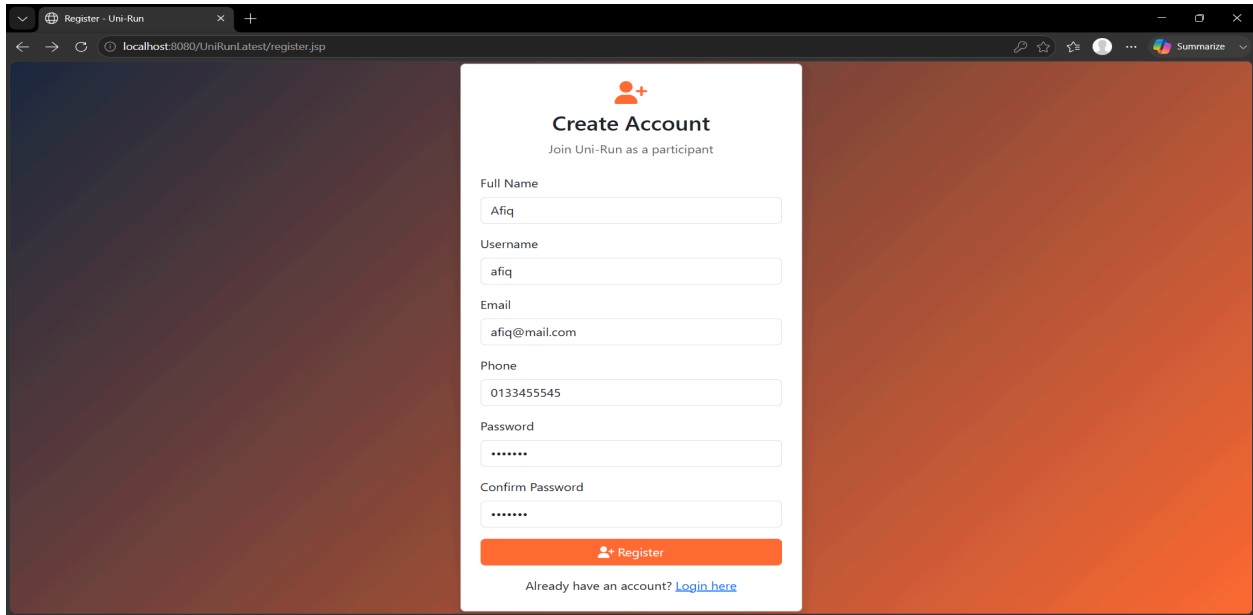
USER MANUAL

1. Participant Interface



Login Page

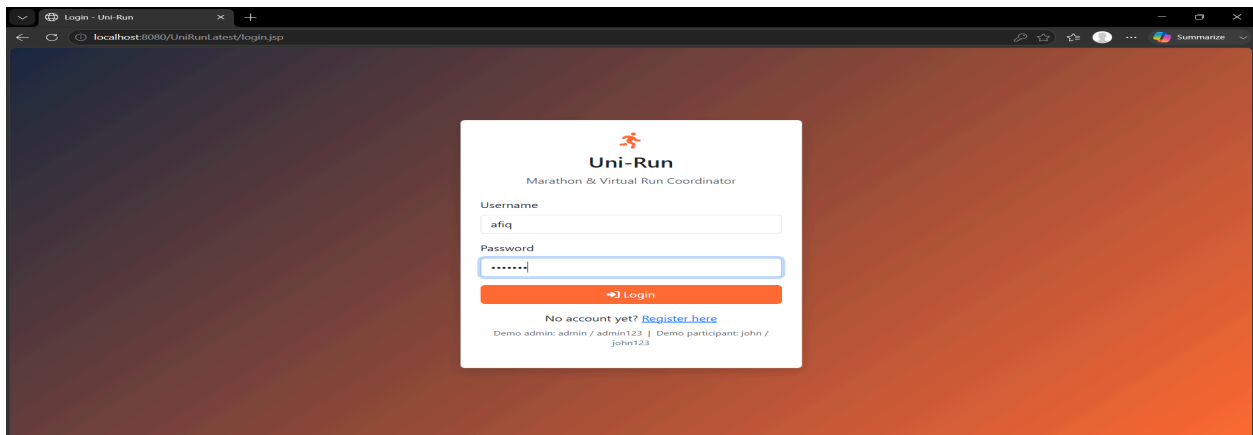
- **Enter Username and Password:** Input your registered username and password into the respective fields.
- **Click Login:** Once the credentials are entered correctly, click the **Login** button to access your dashboard.
- **No Account Yet:** If you don't have an account, click **Register here** to create a new participant account.



The screenshot shows a web browser window with the address bar displaying 'localhost:8080/UniRunLatest/register.jsp'. The page features a central white registration form titled 'Create Account' with a sub-header 'Join Uni-Run as a participant'. The form includes input fields for 'Full Name' (containing 'Afiq'), 'Username' (containing 'afiq'), 'Email' (containing 'afiq@mail.com'), 'Phone' (containing '0133455545'), 'Password' (masked with dots), and 'Confirm Password' (also masked). An orange 'Register' button is positioned below the form, and a link 'Login here' is provided for existing users.

Register Page

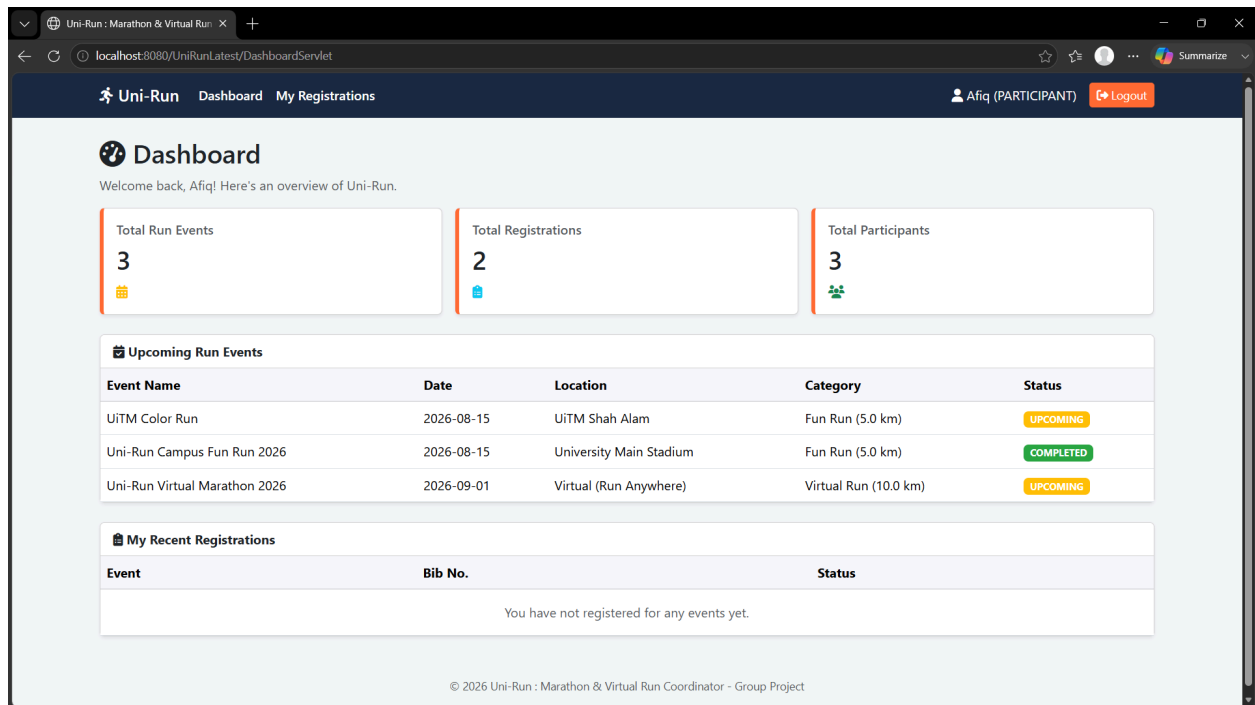
- **Full Name:** Enter your full name.
- **Username:** Choose a unique username.
- **Email:** Enter a valid email address.
- **Phone:** Add your contact number.
- **Password & Confirm Password:** Create and confirm a secure password.
- **Click Register:** Submit the form to complete your registration, then proceed to log in.



The screenshot shows a web browser window with the address bar displaying 'localhost:8080/UniRunLatest/login.jsp'. The page features a central white login form titled 'Uni-Run' with a sub-header 'Marathon & Virtual Run Coordinator'. The form includes input fields for 'Username' (containing 'afiq') and 'Password' (masked with dots). An orange 'Login' button is positioned below the form. A link 'Register here' is provided for users without an account. At the bottom, a footer line reads: 'Demo admin: admin / admin123 | Demo participant: john / john123'.

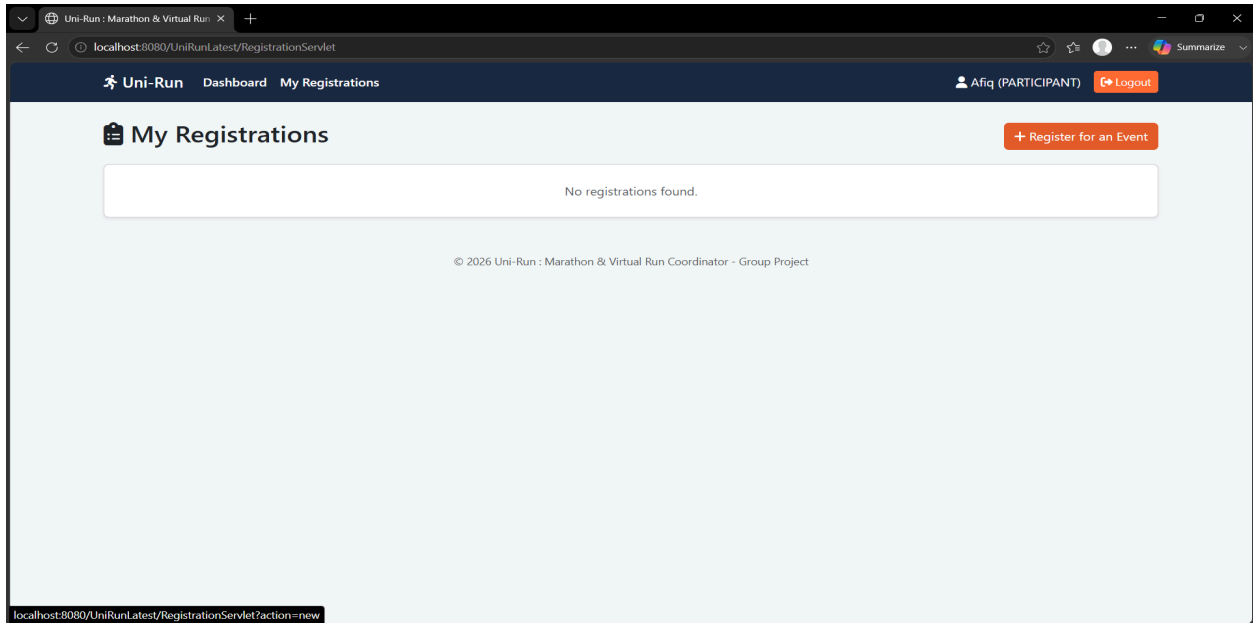
Login Page

- **Enter Username and Password:** Input your registered username and password into the respective fields after successful registration.



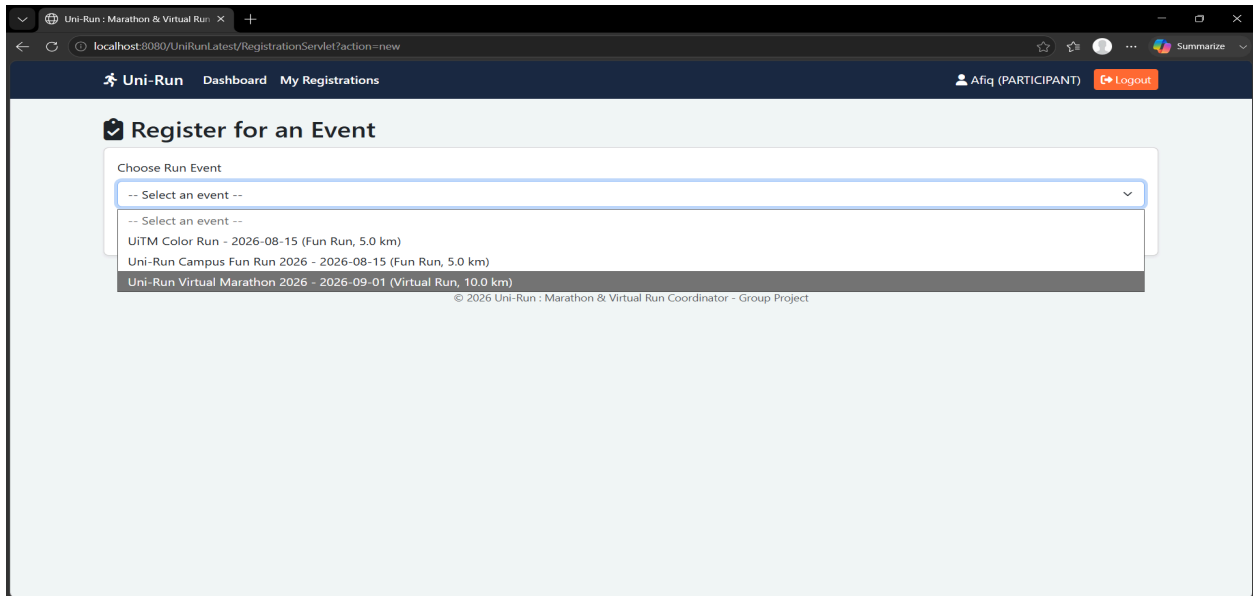
Dashboard

- **View Overview:** Once logged in, the dashboard displays a summary of Total Run Events, Total Registrations, and Total Participants.
- **View Upcoming Run Events:** Browse the list of upcoming events, including event name, date, location, category, and status (e.g., Upcoming, Completed).
- **Track My Recent Registrations:** See the events you've registered for along with your bib number and registration status.



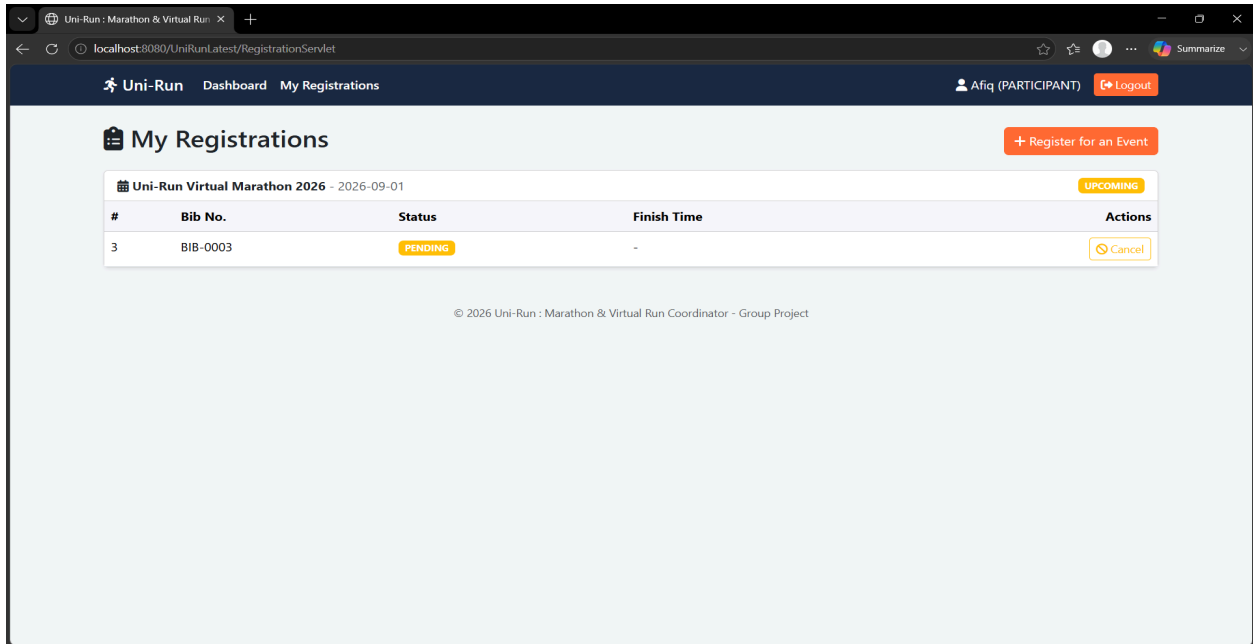
My Registrations

- **View Registrations:** On the **My Registrations** page, view all events you have signed up for, including bib number, status, and finish time (once available).
- **Register for an Event:** Click **Register for an Event** at the top right to sign up for a new run.



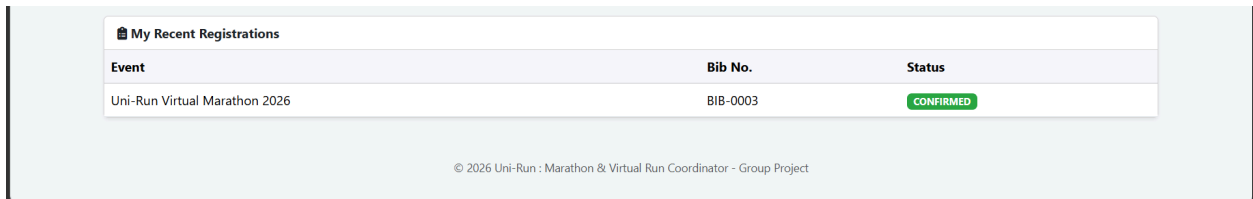
My Registrations

- **Choose an Event:** Select an available event from the dropdown list (e.g., Fun Run or Virtual Run categories).



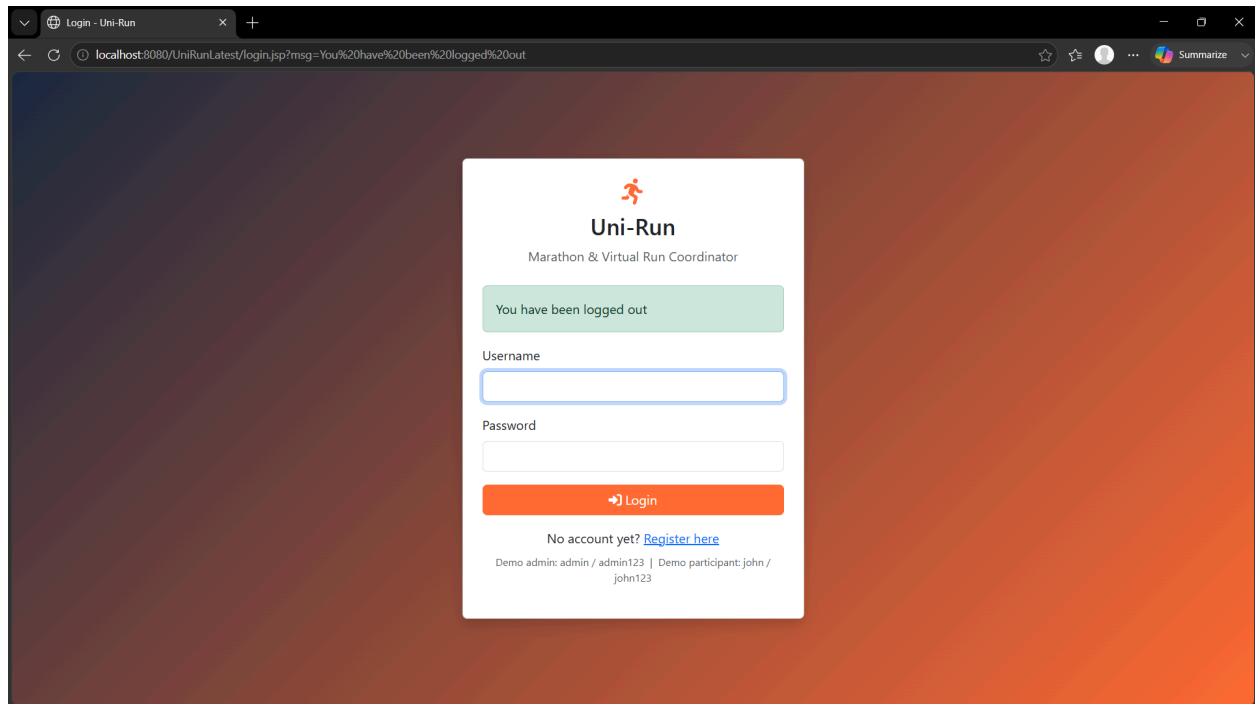
My Registrations

- **Cancel Registration:** If needed, click **Cancel** next to a pending registration to withdraw.
- **Pending Status:** After registering, your entry will show as **Pending** until reviewed.



Registration Approval

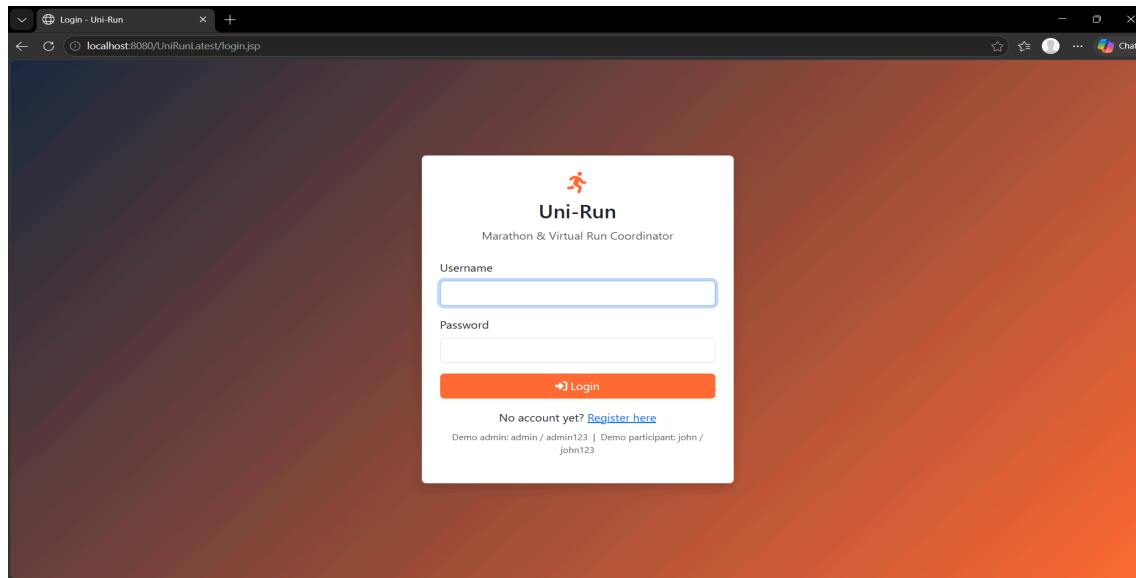
- **Admin Approval:** Once the administrator confirms your registration, the status will update to **CONFIRMED**, and your bib number will be finalized.



Logout

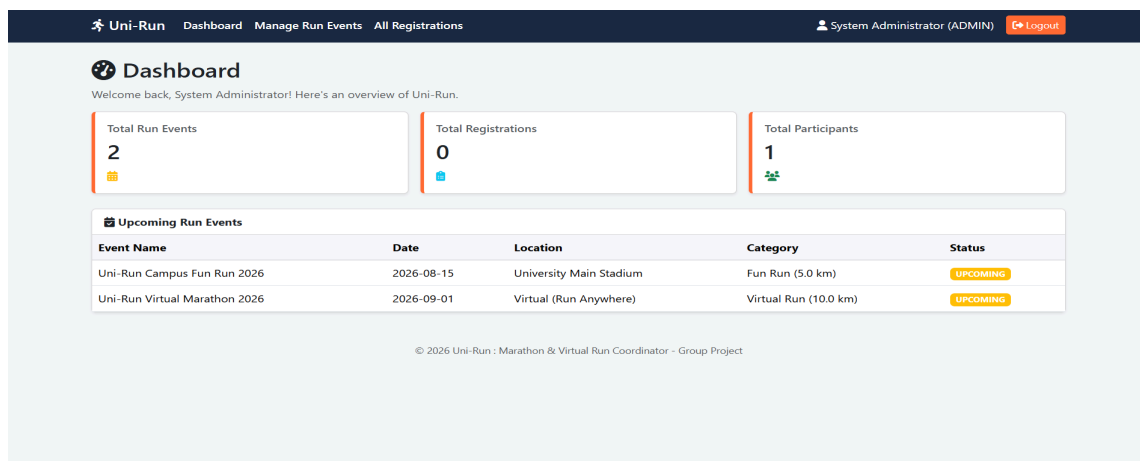
- **Click Logout:** Select the **Logout** button on the top right of the navigation bar to securely sign out of your account.

2. Administrator Interface



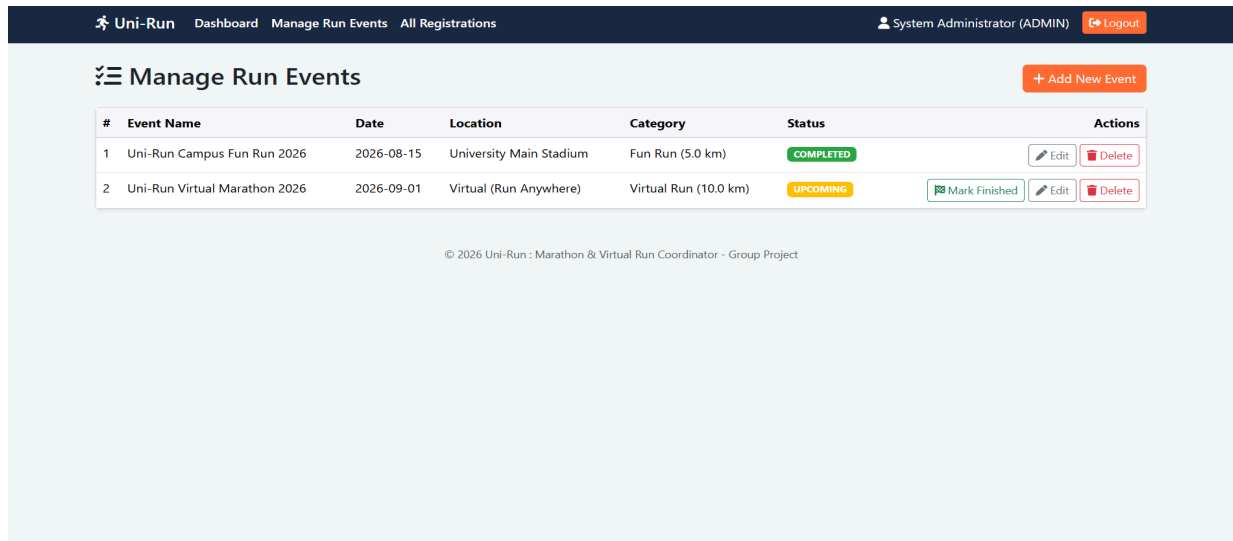
Login Page

- **Enter Admin Credentials:** Log in using the administrator username and password (demo credentials: **admin / admin123**).
- **Click Login:** Access the admin dashboard upon successful login.



Dashboard

- **View System Overview:** The dashboard displays Total Run Events, Total Registrations, and Total Participants at a glance.
- **View Upcoming Run Events:** A summary table lists upcoming events with their date, location, category, and status.



Manage Run Events

- **View All Events:** The **Manage Run Events** page lists all events with details such as name, date, location, category, and status.

+ Add New Run Event

Event Name:

Event Date:

Location:

Category:

Status:

Description:

[Save](#) [Cancel](#)

Add New Event

- **Click Add New Event:** Select the **Add New Event** button at the top right of the Manage Run Events page.
- **Fill in Event Details:** Enter the Event Name, Event Date, Location, Category, Status, and Description.
- **Click Save:** Submit the form to create the new run event, which will then appear in the event list.

Uni-Run

Dashboard

Manage Run Events

All Registrations

System Administrator (ADMIN)

Logout

Edit Run Event

Event Name

Uni-Run Campus Fun Run 2026

Event Date

15/08/2026

Location

University Main Stadium

Category

Fun Run (5.0 km - RM 20.0)

Status

COMPLETED

Description

A fun 5km run open to all students and staff.

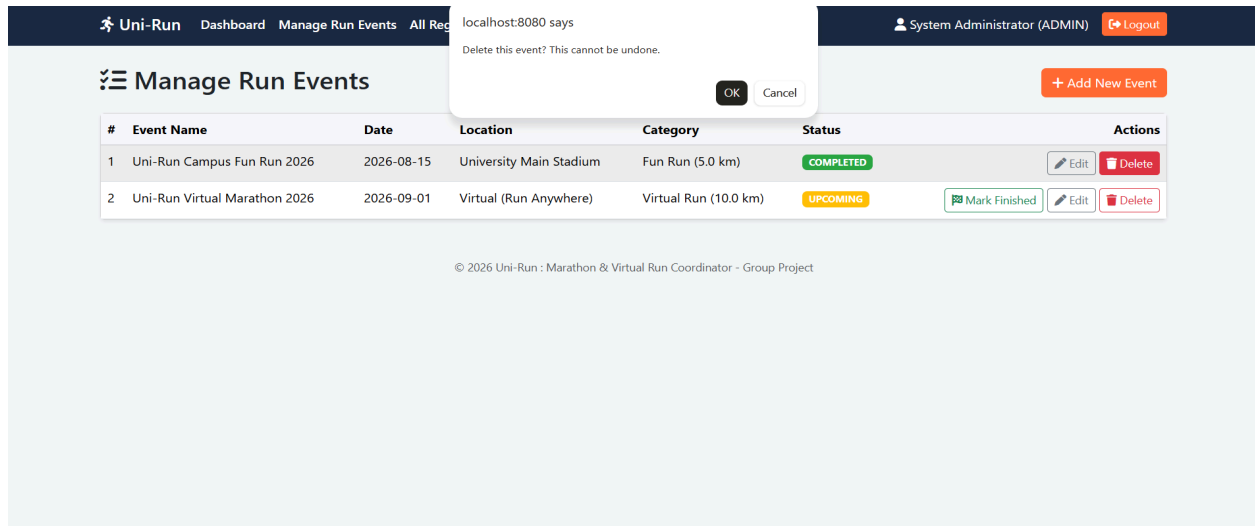
Save

Cancel

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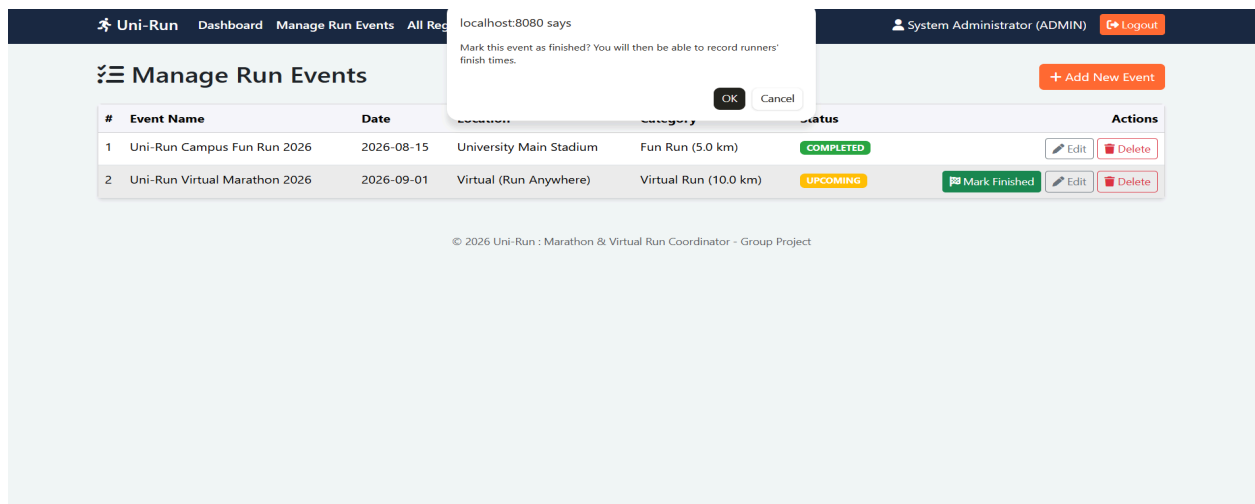
Edit Event

- **Click Edit:** Select **Edit** next to the event you want to update.
- **Update Details:** Modify any of the event fields, such as name, date, location, category, status, or description.
- **Click Save:** Confirm the changes to update the event record.



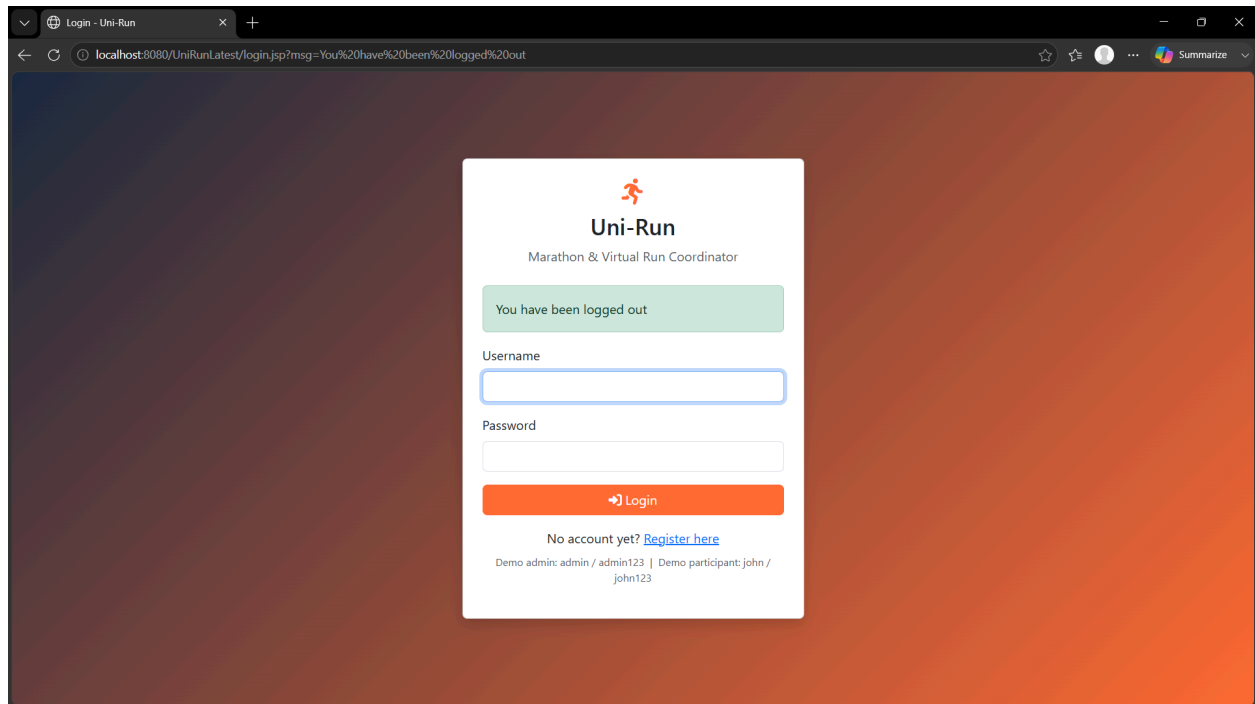
Delete Event

- **Click Delete:** Select **Delete** next to the event you want to remove.
- **Confirm Deletion:** A confirmation prompt will appear asking "Delete this event? This cannot be undone." Click **OK** to proceed or **Cancel** to keep the event.



Mark Event as Finished

- **Click Mark Finished:** Select **Mark Finished** next to an upcoming event once it has taken place.
- **Confirm Action:** A prompt will appear stating "Mark this event as finished? You will then be able to record runners' finish times." Click **OK** to confirm.
- **Status Updates:** The event's status changes to **Completed**, and the finish time field becomes editable for that event's registrations.



Logout

- **Click Logout:** Select the **Logout** button to end the admin session. You will be redirected to the login page with a confirmation message.

CONCLUSION

In conclusion, the development of the Uni-Run: Marathon & Virtual Run Coordinator successfully addresses the critical inefficiencies and administrative bottlenecks associated with managing campus and virtual running events. By transitioning from fragmented manual tracking to a centralized digital platform, the system significantly reduces the labor-intensive processes of registration verification, bib assignment, and race result compilation. Uni-Run provides a seamless and user-friendly experience for both participants and event coordinators, effectively bridging the gap between physical and virtual event management.

From a technical standpoint, the implementation of the Model-View-Controller (MVC) architecture using Java Enterprise Web Technologies ensures that the application is robust, maintainable, and secure. The distinct separation of JavaBeans (Model), JSP (View), and Servlets (Controller) establishes a highly organized framework that effectively mitigates security vulnerabilities through strict session management for user authentication. Furthermore, the comprehensive CRUD modules empower administrators with precise control over event statuses and participant data, ensuring data accuracy and reliability from the initial sign-up phase to the final finish time recording.

Ultimately, Uni-Run delivers substantial long-term value through its dynamic and interactive data visualization capabilities. By offering a centralized dashboard that tracks real-time community-wide metrics, such as cumulative distance covered and demographic breakdowns, event coordinators are equipped with actionable insights. This not only streamlines current event operations but also paves the way for data-driven decision-making, allowing organizers to refine future marketing strategies and foster greater community engagement in future events.